Van-Giang Trinh: Curriculum Vitae

1 Personal



- Name: Van-Giang Trinh (Trịnh Văn Giang in Vietnamese)
- Address: 268 Ly Thuong Kiet, Dien Hong Ward, Ho Chi Minh City, Vietnam
- Email: van-giang.trinh@hcmut.edu.vn
- Google Scholar, Website
- Citizenship: Vietnamese

2 Employment History and Education

2.1 Employment History

- Lecturer, Faculty of Computer Science and Engineering, Ho Chi Minh City University of Technology 09/2025–present
- Postdoc, Lifeware team, Inria Saclay, Palaiseau, France 10/2024–08/2025

  Supervisor: Research Director (DRCE). François Fages
- Postdoc, Laboratoire d'Informatique & Systèmes, Aix-Marseille University 04/2022–09/2024

  Supervisor: Prof. Belaid Benhamou
- Part-time Researcher, School of Information Science, Japan Advanced Institute of Science and Technology 01/2022–03/2022
- Doctoral Student, School of Information Science, Japan Advanced Institute of Science and Technology 10/2018–12/2021
- Research Student, School of Information Science, Japan Advanced Institute of Science and Technology 10/2017–09/2018
- Research Assistant, Faculty of Computer Science and Engineering, Ho Chi Minh City University of Technology 06/2016–06/2017
- Internship, School of Information Science, Japan Advanced Institute of Science and Technology 10/2015–11/2015

2.2 Education

- **Ph.D.**, Information Science, Japan Advanced Institute of Science and Technology, Japan 10/2018–12/2021

  Supervisor: Prof. Kunihiko Hiraishi

  Thesis Title: On attractor detection and optimal control of Boolean networks

  Jury: Prof. Kunihiko Hiraishi, Prof. Tatsuya Akutsu, Assoc. Prof. Koichi Kobayashi, Prof. Mineo Kaneko, and Assoc. Prof. Daisuke Ishii
- **Master of Science**, Computer Science, Ho Chi Minh City University of Technology, Vietnam 08/2014–04/2017

  Supervisor: Assoc. Prof. Quan Thanh Tho
- **Bachelor of Science**, Computer Science, Ho Chi Minh City University of Technology, Vietnam 09/2009–04/2014

Supervisor: Assoc. Prof. Quan Thanh Tho

- **High School**, Le Loi High School, Tho Xuan, Thanh Hoa, Vietnam 09/2006–05/2009

3 Research

3.1 Research Interests

My research interests lie at the intersection of theoretical computer science, symbolic artificial intelligence, and computational systems biology. I focus on formal frameworks such as Boolean networks, Petri nets, answer set programming, and abstract argumentation, exploring their interconnections and their applications to the modeling, analysis, and control of biological systems.

3.2 Awards

1. Outstanding Performance Award, Japan Advanced Institute of Science and Technology 12/2021
2. Japanese Government (Monbukagakusho: MEXT) Scholarship, Japan Advanced Institute of Science and Technology 10/2017–09/2021
3. JAIST President Award, Japan Advanced Institute of Science and Technology 09/2019

3.3 Publications

3.3.1 Submitted

1. Samuel Pastva, & **Van-Giang Trinh**. (2025). BAss: Symbolic reasoning in abstract dialectical frameworks. In ***International Symposium on Formal Methods***. (under review)

2. **Van-Giang Trinh**, Belaid Benhamou, Sylvain Soliman, & François Fages. (2025). On the Boolean network theory of Datalog$^{\neg}$. ***Theory and Practice of Logic Programming***. (under review) https://doi.org/10.48550/arXiv.2504.15417

3.3.2 International Journals (top venues in bold)

1. **Van-Giang Trinh**, Kyu Hyong Park, Samuel Pastva, & Jordan C Rozum. (2025). Mapping the attractor landscape of Boolean networks via biobalm. ***Bioinformatics***, *41*(5). (Scimago Q1, IF 4.4, Google Scholar top 1/20) https://doi.org/10.1093/bioinformatics/btaf280

2. **Van-Giang Trinh**, Belaid Benhamou, & Sylvain Soliman. (2023b, September). Trap spaces of Boolean networks are conflict-free siphons of their Petri net encoding. ***Theoretical Computer Science***, *971*, 114073. (Scimago Q2, IF 1.1, CORE Rank A, Google Scholar top 5/20) https://doi.org/10.1016/j.tcs.2023.114073

3. Tarek Khaled, Belaid Benhamou, & **Van-Giang Trinh**. (2023, July). Using answer set programming to deal with Boolean networks and attractor computation: Application to gene regulatory networks of cells. ***Annals of Mathematics and Artificial Intelligence***, 1–38. (Scimago Q3, IF 1.2, CORE Rank C) https://doi.org/10.1007/s10472-023-09886-7

4. **Van-Giang Trinh**, Belaid Benhamou, Thomas Henzinger, & Samuel Pastva. (2023a, June). Trap spaces of multi-valued networks: Definition, computation, and applications. ***Bioinformatics***, *39*(Supplement_1), i513–i522. (Scimago Q1, IF 5.8, Google Scholar top 1/20) https://doi.org/10.1093/bioinformatics/btad262

5. **Van-Giang Trinh**, & Kunihiko Hiraishi. (2020b). On attractor detection and optimal control of deterministic generalized asynchronous random Boolean networks. ***IEEE/ACM Transactions on Computational Biology and Bioinformatics***, *19*(3), 1794–1806. (Scimago Q2, IF 4.5, CORE Rank C, Google Scholar top 7/20) https://doi.org/10.1109/TCBB.2020.3043785

6. **Van-Giang Trinh**, Tatsuya Akutsu, & Kunihiko Hiraishi. (2020). An FVS-based approach to attractor detection in asynchronous random Boolean networks. ***IEEE/ACM Transactions on Computational Biology and Bioinformatics***, *19*(2), 806–818. (Scimago Q2, IF 4.5, CORE Rank C, Google Scholar top 7/20) https://doi.org/10.1109/TCBB.2020.3028862

7. **Van-Giang Trinh**, & Kunihiko Hiraishi. (2020c). A study on attractors of generalized asynchronous random Boolean networks. *IEICE TRANSACTIONS on Fundamentals of Electronics, Communications and Computer Sciences*, *103*(8), 987–994. (Scimago Q3, IF 0.3, CORE Rank C) https://doi.org/10.1587/transfun.2019EAP1163

8. **Van-Giang Trinh**, Le Ngoc Kim Khanh, Bang Ngoc Bao Tam, Tram Loi Quan, Bui Hoai Thang, & Quan Thanh Tho. (2016). Modelling and congestion detection of wireless sensor networks: A concurrent-based approach using coloured Petri nets. *International Journal of Applied Information Systems*, *11*(7), 1–9. https://doi.org/10.5120/ijais2016451629

9. Pham Hong Long, **Van-Giang Trinh**, Dinh Hoang Mai, Mai Phuong Nam, Quan Thanh Tho, & Ngo Quang Hung. (2014). Assisting students in finding bugs and their locations in programming solutions. *International Journal of Quality Assurance in Engineering and Technology Education (IJQAETE)*, *3*(2), 12–27. https://doi.org/10.4018/ijqaete.2014040102

3.3.3 International Conferences (top venues in bold)

1. **Van-Giang Trinh**, Sylvain Soliman, François Fages, & Belaid Benhamou. (2025, September). On the trap space semantics of normal logic programs. In ***International Conference on Logic Programming***. (regular paper, CORE Rank A, top-one in logic programming)

2. Mohimenul Kabir*, **Van-Giang Trinh***, Samuel Pastva*, & Kuldeep S. Meel. (2025, August). Scalable counting of minimal trap spaces and fixed points in Boolean networks. In ***International Conference on Principles and Practice of Constraint Programming***. (regular paper, * = equal contribution, AR 39%, CORE Rank A, top-one in constraint programming) https://doi.org/10.4230/LIPIcs.CP.2025.17

3. **Van-Giang Trinh**, Samuel Pastva, Jordan C Rozum, Kyu Hyong Park, & Réka Albert. (2025, June). On the number of asynchronous attractors in AND-NOT Boolean networks. In ***International Workshop on Cellular Automata and Discrete Complex Systems***. (exploratory paper) https://doi.org/10.48550/arXiv.2503.19147

4. **Van-Giang Trinh**, Belaid Benhamou, & Vincent Risch. (2025, February). Graphical analysis of abstract argumentation frameworks via Boolean networks. In *International Conference on Agents and Artificial Intelligence* (pp. 745–756). (regular paper, AR 23.09%, CORE Rank B) https://doi.org/10.5220/0013346400003890

5. Quang-Anh Nguyen, Thu-Trang Pham, Thi-Hai-Yen Vuong, **Van-Giang Trinh**, & Ha-Thanh Nguyen. (2025, February). Detecting misleading information with LLMs and explainable ASP. In *International Conference on Agents and Artificial Intelligence* (pp. 1327–1334). (short paper, CORE Rank B) https://doi.org/10.5220/0013357400003890

6. **Van-Giang Trinh**, Belaid Benhamou, Sylvain Soliman, & François Fages. (2024, October). Graphical conditions for the existence, unicity and number of regular models. In ***International Conference on Logic Programming*** (pp. 175–187). (regular paper, CORE Rank A, top-one in logic programming) https://doi.org/10.4204/EPTCS.416.16

7. **Van-Giang Trinh**, Belaid Benhamou, Samuel Pastva, & Sylvain Soliman. (2024, February). Scalable enumeration of trap spaces in Boolean networks via answer set programming. In ***Annual AAAI Conference on Artificial Intelligence*** (p. 10714-10722). (regular paper, **oral presentation**, AR 23.8% for the overall conference and 2.3% for oral presentation, CORE Rank A*, top-one in general AI) https://doi.org/10.1609/aaai.v38i9.28943

8. **Van-Giang Trinh**, Belaid Benhamou, & Sylvain Soliman. (2023a, August). Efficient enumeration of fixed points in complex Boolean networks using answer set programming. In ***International Conference on Principles and Practice of Constraint Programming*** (pp. 35:1–35:19). (regular paper, AR 40%, CORE Rank A, top-one in constraint programming) https://doi.org/10.4230/LIPIcs.CP.2023.35

9. **Van-Giang Trinh**, Belaid Benhamou, Thomas Henzinger, & Samuel Pastva. (2023b, July). Trap spaces of multi-valued networks: Definition, computation, and applications. In ***The 31st Annual Intelligent Systems For Molecular Biology and the 22nd Annual European Conference on Computational Biology***. Oxford University Press. (regular paper, AR 17.9%, CORE Rank A, top-one in computational biology) https://doi.org/10.1093/bioinformatics/btad262

10. **Van-Giang Trinh**, Kunihiko Hiraishi, & Belaid Benhamou. (2022, August). Computing attractors of large-scale asynchronous Boolean networks using minimal trap spaces. In ***ACM International Conference on Bioinformatics, Computational Biology and Health Informatics*** (pp. 1–10). ACM. (regular paper, AR 29%, flagship conference of the ACM SIGBio) https://doi.org/10.1145/3535508.3545520

11. **Van-Giang Trinh**, Belaid Benhamou, Kunihiko Hiraishi, & Sylvain Soliman. (2022, August). Minimal trap spaces of logical models are maximal siphons of their Petri net encoding. In *International Conference on Computational Methods in Systems Biology* (pp. 158–176). Springer. (regular paper, AR 65%) https://doi.org/10.1007/978-3-031-15034-0_8

12. **Van-Giang Trinh**, & Kunihiko Hiraishi. (2021, October). An improved method for finding attractors of large-scale asynchronous Boolean networks. In *IEEE International Conference on Computational Intelligence in Bioinformatics and Computational Biology* (pp. 1–9). IEEE. (regular paper, AR 51%, CORE Rank C) https://doi.org/10.1109/cibcb49929.2021.9562947

13. **Van-Giang Trinh**, & Kunihiko Hiraishi. (2019, June). Algorithms for finding attractors of generalized asynchronous random Boolean networks. In *12th Asian Control Conference* (pp. 67–72). IEEE. Retrieved from `http://ieeexplore.ieee.org/document/8765169` (regular paper, AR 73%)

14. Le Ngoc Kim Khanh, **Van-Giang Trinh**, Bui Hoai Thang, & Quan Thanh Tho. (2017, April). Probabilistic modelling for congestion detection on wireless sensor networks. In *International Conference on Control, Decision and Information Technologies* (pp. 0190–0195). IEEE. (regular paper, AR 48%, CORE Rank C) https://doi.org/10.1109/CoDIT.2017.8102589

15. **Van-Giang Trinh**, Nguyen Duc Khoan, Nguyen Duy Khuong, Vu Phu Thuc, & Quan Thanh Tho. (2016, September). Fast-and-Fit: An intelligent auto-pricing system for airlines travel agencies. In *SAI Intelligent Systems Conference* (pp. 853–865). Springer. (regular paper) https://doi.org/10.1007/978-3-319-56994-9_58

16. **Van-Giang Trinh**, Kunihiko Hiraishi, & Quan Thanh Tho. (2016, July). Modeling and analysing Boolean networks by coloured Petri nets. *IEICE Proceedings Series*, *61*(4447). (regular paper) https://doi.org/10.34385/proc.61.4447

17. Bao Trung Pham Duy, **Van-Giang Trinh**, Le Dinh Thuan, & Quan Thanh Tho. (2015, October). Reusing symbolic observation graph for efficient model checking. In *International Conference on Knowledge and Systems Engineering* (pp. 250–255). IEEE. (regular paper) https://doi.org/10.1109/kse.2015.44

3.3.4 International Workshops

1. **Van-Giang Trinh**, & Kunihiko Hiraishi. (2020a, December). An efficient method for approximating attractors in large-scale asynchronous Boolean models. In *International Workshop on Biological Network Analysis and Integrative Graph-Based Approaches, in conjunction with IEEE International Conference on Bioinformatics and Biomedicine* (pp. 1820–1826). IEEE. (workshop paper, AR 60%) https://doi.org/10.1109/bibm49941.2020.9313230

3.3.5 Technical Reports

1. **Van-Giang Trinh**, & Belaid Benhamou. (2024). Static analysis of logic programs via Boolean networks. *CoRR*, *abs/2407.09015*. https://doi.org/10.48550/ARXIV.2407.09015

2. **Van-Giang Trinh**, Belaid Benhamou, & Loïc Paulevé. (2024). mpbn: a simple tool for efficient edition and analysis of elementary properties of Boolean networks. *CoRR*, *abs/2403.06255*. https://doi.org/10.48550/ARXIV.2403.06255

3.3.6 Other Publications

1. F. Kordon, H. Garavel, L. M. Hillah, F. Hulin-Hubard and G. Chiardo, A. Hamez, L. Jezequel, A. Miner, J. Meijer, E. Paviot-Adet, D. Racordon, C. Rodriguez, C. Rohr, J. Srba, Y. Thierry-Mieg, **Van-Giang Trinh**, & K. Wolf. (2016, June). *Complete Results for the 2016 Edition of the Model Checking Contest*. Retrieved from `https://mcc.lip6.fr/2016/index.php`

3.4 Talks and Seminars

1. PTIT seminar (Hanoi, Vietnam), Scalable enumeration of trap spaces in Boolean networks via answer set programming (28/08/2025)

2. HUS seminar (Hanoi, Vietnam), Scalable enumeration of trap spaces in Boolean networks via answer set programming (27/08/2025)

3. Joint seminar of the Department of Discrete Mathematics, Probabilily Theory and Statistics, and the ICRTM (Hanoi, Vietnam), Connecting Abstract Argumentation Frameworks and Boolean Networks (12/08/2025)

4. MLLab seminar (Hanoi, Vietnam), Scalable enumeration of trap spaces in Boolean networks via answer set programming (07/08/2025)

5. LIRICA Seminar (St Charles, Marseille), On the Trap Space Semantics of Normal Logic Programs (26/06/2025)

6. LIRICA Seminar (St Charles, Marseille), Graphical conditions for the existence, unicity and number of regular models (25/11/2024)

7. MABioS seminar (Luminy, Marseille), Trap spaces of multi-valued networks: definition, computation, and applications (30/09/2024)

8. Seminar on Mathematical Foundations for Computer Science (Hanoi, Vietnam), Trap spaces of Boolean networks are conflict-free siphons of their Petri net encoding (20/08/2024)

9. UET seminar (Hanoi, Vietnam), Scalable enumeration of trap spaces in Boolean networks via answer set programming (19/08/2024)

10. CSE-HCMUT seminar (Ho Chi Minh city, Vietnam), Scalable enumeration of trap spaces in Boolean networks via answer set programming (12/08/2024)

11. Journées annuelles 2024 du GT Bioss (LIP6, Paris), Trap spaces of multi-valued networks: definition, computation, and applications (27/05/2024)

12. LIRICA Seminar (St Charles, Marseille), Scalable enumeration of trap spaces in Boolean networks via answer set programming (25/03/2024)
13. Lifeware Seminar (INRIA Saclay), Trap spaces of Boolean networks are conflict-free siphons of their Petri net encoding (05/12/2023)
14. IRISA Seminar (IRISA, Rennes), Efficient enumeration of fixed points in complex Boolean networks using answer set programming (30/11/2023)
15. LIRICA Seminar (St Charles, Marseille), Efficient enumeration of fixed points in complex Boolean networks using answer set programming (20/11/2023)
16. Demi Journées du Pôle Calcul on Artificial Intelligence (Luminy, Marseille), Efficient enumeration of fixed points in complex Boolean networks using answer set programming (15/06/2023)
17. CANA Seminar (Luminy, Marseille), Trap spaces of Boolean networks are conflict-free siphons of their Petri net encoding (30/05/2023)
18. Journées BioLogique BIOSS/CAVIAR (LIP6, Paris), An approach based on ASP and Petri nets for the calculation of attractors in Boolean networks (25/05/2023)
19. Journées Scientifiques du LIS (Carry le Rouet), Efficient enumeration of minimal trap spaces in large-scale Boolean networks of gene networks (23/05/2023)
20. LIRICA Seminar (St Charles, Marseille), Minimal trap spaces of Boolean models are maximal siphons of their Petri net encoding (17/10/2022)
21. LIRICA Seminar (Virtual), An FVS-based approach to attractor detection in asynchronous Boolean networks (28/06/2021)

4 Teaching

4.1 Courses

- Exercises on Graph Theory (graduate course, teaching assistant)
  - Japan Advanced Institute of Science and Technology: 06/2021–08/2021
- Functional Programming (graduate course, teaching assistant)
  - Japan Advanced Institute of Science and Technology: 10/2020–12/2020
- Introduction to Computer Programming (undergraduate course, visiting lecturer)
  - Ho Chi Minh City University of Technology: 01/2017–06/2017
- Principles of Programming Languages (undergraduate course, teaching assistant)
  - Ho Chi Minh City University of Technology: 09/2013–01/2014, 09/2014–01/2015, 09/2015–01/2016
- Object-Oriented Programming (undergraduate course, teaching assistant)
  - Ho Chi Minh City University of Technology: 09/2015–01/2016
- Principles of Programming Languages (undergraduate course, visiting lecturer)
  - Ho Chi Minh City University of Natural Resources and Environment: 01/2015–05/2015
- Data Structures and Algorithms (undergraduate course, teaching assistant)
  - Ho Chi Minh City University of Technology: 09/2012–01/2013, 01/2015–05/2015
- Programming Fundamentals (undergraduate course, teaching assistant)
  - Ho Chi Minh City University of Technology: 01/2015–05/2015

5 Professional Services

5.1 Program Committee Memberships

- International Conference on Principles of Knowledge Representation and Reasoning 2026
- International Workshop on Nonmonotonic Reasoning 2026
- International Workshop on Nonmonotonic Reasoning 2025
- Workshop on Answer Set Programming and Other Computing Paradigms 2025

5.2 Journal Reviewers

- SIAM Journal on Applied Dynamical Systems (2023—1 paper)
- IEEE/ACM Transactions on Computational Biology and Bioinformatics (2022—1 paper, 2023—1 paper)

5.3 Conference Reviewers

- International Conference on Logic Programming 2025 (subreviewer—1 paper)
- International Conference on Computational Methods in Systems Biology 2025 (subreviewer—1 paper)

6 References

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